United States Patent	12408296
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Zhou; Feng et al.

Power device embedded printed circuit board-cold plate assemblies with low interfacial thermal and mechanical stresses and methods of making the same

Abstract

A highly integrated power electronics embedded PCB-cold plate assembly includes a cold plate, a power electronics embedded printed circuit board (PCB), and a low thermal resistance dielectric layer sandwiched between the cold plate and the power electronics embedded PCB. The power electronics embedded PCB is bonded to the cold plate via the low thermal resistance dielectric layer to form highly integrated power electronics embedded PCB—cold plate assembly is formed. And in one example, the low thermal resistance dielectric layer sandwiched between and directly bonded to the cold plate and the power electronics embedded PCB.

Inventors: **Zhou; Feng (Ann Arbor, MI), Fan; Tianzhu (Houston, TX), Liu; Yanghe (Ann Arbor, MI)**

Applicant: **Toyota Motor Engineering & Manufacturing North America, Inc.** (Plano, TX)

Family ID: **92460406**

Assignee: **Toyota Motor Engineering & Manufacturing North America, Inc. (Plano, TX)**

Appl. No.: **18/114622**

Filed: **February 27, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240292564 A1	Aug. 29, 2024

Publication Classification

Int. Cl.: **H05K7/20** (20060101); **H05K1/02** (20060101)

U.S. Cl.:

CPC **H05K7/20254** (20130101); **H05K1/0272** (20130101); **H05K7/20927** (20130101);

Field of Classification Search

CPC: H05K (7/20254); H05K (7/20309); H05K (7/20336); H05K (7/20509); H05K (7/2089); H05K (7/20927); H05K (7/20936); H05K (1/0203); H05K (1/0204); H05K (1/0207); H05K (1/0209); H05K (1/0272); H05K (1/0298); H05K (1/05); H05K (1/183); H05K (1/185); H01L (23/3121); H01L (23/367); H01L (23/3675); H01L (23/3735); H01L (23/427); H01L (23/473); H01L (21/4803); H01L (21/4881); H01L (2224/04105)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2011/0163457	12/2010	Mohan	257/E23.079	H01L 23/49822
2014/0175633	12/2013	Lin	257/712	H01L 23/36
2014/0318830	12/2013	Robert	N/A	N/A
2018/0366400	12/2017	Mohn et al.	N/A	N/A
2019/0131208	12/2018	Feichtinger et al.	N/A	N/A
2022/0013431	12/2021	Zhou	N/A	H01L 23/3677
2022/0053634	12/2021	Zhou	N/A	H01L 23/427
2022/0065549	12/2021	Zhou	N/A	F28D 15/0275
2022/0070996	12/2021	Esler	N/A	H05K 7/20509
2022/0141951	12/2021	Zhou	361/700	H05K 1/185
2022/0157693	12/2021	Zhou	N/A	H01L 25/50
2022/0167528	12/2021	Joshi	N/A	H05K 7/20936
2022/0192002	12/2021	Zhou	N/A	H05K 3/30
2022/0287209	12/2021	Reiter et al.	N/A	N/A
2023/0022829	12/2022	Zhou	N/A	H01L 23/3735
2023/0238301	12/2022	Miorini	257/774	H01L 24/24
2024/0251507	12/2023	Ewanchuk	N/A	H05K 1/115

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017228623	12/2016	JP	N/A

OTHER PUBLICATIONS

JP-2017228623-A English Translation (Year: 2017). cited by examiner

---

Primary Examiner: Jalali; Amir A

Attorney, Agent or Firm: Darrow Mustafa PC

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to printed circuit boards, and particularly to printed circuit boards with power devices embedded therein.

BACKGROUND

(2) Printed circuit boards (PCBs) are typically used for mechanical support and electrical connection of electronic components using conductive pathways of copper sheets laminated onto a non-conductive substrate. And multi-layer PCBs provide higher capacity and/or density of electronic components in a smaller footprint by incorporating two or more layers. However, the design and/or manufacture of multilayer PCBs can be difficult.

(3) The present disclosure addresses issues related to the manufacture of multi-layer PCBs and other issues related to multi-layer PCBs.

SUMMARY

(4) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(5) In one form of the present disclosure, an assembly includes a cold plate, a power electronics embedded printed circuit board (PCB), and a low thermal resistance (LTR) dielectric layer sandwiched between the cold plate and the power electronics embedded PCB such that the power electronics embedded PCB is bonded to the cold plate via the LTR dielectric layer and forms a highly integrated power electronics embedded PCB—cold plate assembly is formed.

(6) In another form of the present disclosure, a highly integrated power electronics embedded PCB—cold plate assembly includes a cold plate, a power electronics embedded PCB, and a LTR dielectric layer sandwiched between and directly bonded to the cold plate and the power electronics embedded PCB.

(7) In still another form of the present disclosure, a method includes laminating a LTR dielectric layer onto a cold plate fabrication panel, the cold plate fabrication panel including a plurality of individual cold plate units, bonding a power electronics embedded PCB fabrication panel to the LTR dielectric layer using high temperature and high pressure and forming an integrated power electronics PCB—cold plate fabrication panel, and cutting the integrated power electronics PCB—cold plate fabrication panel into a plurality of individual highly integrated power electronics embedded PCB—cold plate units.

(8) Further areas of applicability and various methods of enhancing the above technology will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present teachings will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. **1**A shows a perspective view of a multi-layer PCB for a highly integrated power electronics embedded PCB—cold plate assembly according to the teachings of the present disclosure;

(3) FIG. **1**B shows a perspective for a power layer of the multi-layer PCB in FIG. **1**A:

(4) FIG. **2** shows a side view of a highly integrated power electronics embedded PCB—cold plate assembly according to the teachings of the present disclosure;

(5) FIG. **3**A shows a thermal simulation heat map for the highly integrated power electronics embedded PCB—cold plate assembly in FIG. **2**;

(6) FIG. **3**B shows a normal stress simulation map for a bonding interface of the highly integrated

power electronics embedded PCB—cold plate assembly in FIG. **2**;
(7) FIG. **3**C shows a shear stress simulation map for the bonding interface of the highly integrated power electronics embedded PCB—cold plate assembly in FIG. **2**;
(8) FIG. **4**A shows a perspective view of a cold plate fabrication panel according to the teachings of the present disclosure;
(9) FIG. **4**B shows a perspective view of an insulated metal substrate (IMS) deposited on the cold plate fabrication panel in FIG. **4**A;
(10) FIG. **4**C shows a perspective view of a highly integrated power electronics embedded PCB fabrication panel to be bonded to the cold plate fabrication panel in FIG. **4**B;
(11) FIG. **4**D shows a side view of the highly integrated power electronics embedded PCB fabrication panel bonded to the LTR dielectric layer deposited on the cold plate fabrication panel in FIG. **4**C;
(12) FIG. **4**E shows a side cross-section view of section **4**E in FIG. **4**D; and
(13) FIG. **5** shows a flow chart for a method of manufacturing a highly integrated power electronics embedded PCB—cold plate assembly according to the teachings of the present disclosure.
(14) It should be noted that the figures set forth herein are intended to exemplify the general characteristics of the methods, devices, and systems among those of the present technology, for the purpose of the description of certain aspects. The figures may not precisely reflect the characteristics of any given aspect and are not necessarily intended to define or limit specific forms or variations within the scope of this technology.
DETAILED DESCRIPTION
(15) The present disclosure provides highly integrated power electronics embedded PCB—cold plate assemblies with desired heat transfer, and reduced normal and shear stresses, between a cold plate and a power electronics embedded PCB bonded thereto. As used herein, the phrase “highly integrated power electronics embedded PCB” refers to a single multi-layer PCB module or unit with two or more power semiconductor devices (also referred to herein simply as “power device”), control/drive/protection electronic circuitry, and passive components, embedded therein. Also, as used herein, the phrase “power device” refers to a semiconductor device used as a switch or rectifier in power electronics. The cold plate can be fluid cooled such that temperatures of the one or more power device—substrate assemblies during operation remain below a predefined temperature. In addition, the multi-layer PCB with the one or more power device—substrate assemblies embedded therein is electrically isolated from the cold plate via a low thermal resistance dielectric layer.
(16) Referring now to FIGS. **1**A-**1**B, a perspective cross-sectional view of a multi-layer PCB **10** is shown in FIG. **1**A and an isolated perspective cross-sectional view of a power layer **110** of the multi-layer PCB **10** is shown in FIG. **1**B. The multi-layer PCB **10** includes a plurality of dielectric layers **100** and a plurality of the power layers **110**. The power layers **110** include a dielectric material **112** and a conductive material **114**. The dielectric layers **100** include the dielectric material **112** and conductive vias **114***v* that provide electrical communication or pathways between adjacent power layers **110**. Stated differently, the power layers **110** include conductive (e.g., copper) patterns and the dielectric layers **100** include conductive (e.g., copper) pathways that connect the conductive patterns **114** such that the multi-layer PCB **10** functions and/or operates as desired.
(17) In some variations, the power layers **110** are formed from a glass reinforced epoxy laminate dielectric material (e.g., FR-4), or other dielectric material, with the conductive material **114** embedded therein. In other variations, the power layers **110** are 3D printed using a dielectric material ink to form the dielectric material **112**, with the conductive material **114** embedded therein. And in at least one variation, the conductive material **114** is also 3D printed with a conductive material ink. Non limiting examples of dielectric material inks are inks that include UV-curable dielectric materials such as UV-curable acrylated monomer selected from one or more of an acrylate epoxy, an acrylate polyester, an acrylate urethane, and an acrylate silicone, among others.

And non-limiting examples of conductive material inks are inks that include silver nanoparticles and/or graphene nanosheets, among others. It should be understood that the dielectric layers **100** can also be formed from a glass reinforced epoxy laminate dielectric material (e.g., FR-4), or other dielectric material, or 3D printed using a dielectric material ink.

(18) The dielectric layers **100** and the power layers **110** have a predefined average thickness (z-direction). For example, in some variations, the predefined average thickness is between about 50 micrometers (μm) and about 250 μm, for example, between about 75 μm and about 200 μm. And in at least one variation, the predefined thickness is between about 75 μm and about 150 μm, for example, between about 80 μm and about 120 μm.

(19) Referring to FIG. **2**, a side view of a highly integrated power electronics embedded PCB—cold plate assembly **2** according to one form of the present disclosure is shown. The highly integrated power electronics embedded PCB—cold plate assembly **2** includes a power electronics embedded PCB **20** formed from the multi-layer PCB **10** with power device—substrate assemblies **30** embedded therein, and a cold plate **40**. As used herein, the phrase power device—substrate assembly refers to a power device (e.g., a MOSFET power device) attached or bonded to a substrate. In some variations, the substrate is a copper-graphite substrate with a graphite core embedded within a shell of copper. And in such variations, the power device is bonded to the copper shell, e.g., via silver sintering.

(20) In some variations, the power device—substrate assemblies **30** are totally embedded in the multi-layer PCB **10**, i.e., one or more dielectric layers **100** and/or one or more power layers **110** are above (+z direction) the power device—substrate assemblies **30**, and one or more dielectric layers **100** and/or one or more power layers **110** are below (−z direction) the power device—substrate assemblies **30**. In other variations, the power device—substrate assemblies **30** form at least a portion of the lower surface **22** of the power electronics embedded PCB **20**. For example, a lower surface **32** of one or more power device—substrate assemblies **30** is aligned or coplanar with the lower surface **22** of the power electronics embedded PCB **20** (not shown) such that the lower surfaces **22**, **32** of the power electronics embedded PCB **20** and power device(s), respectively, are bonded to the cold plate via a low thermal resistance dielectric layer as described below.

(21) The power electronics embedded PCB **20** is bonded to the cold plate **40** such that the power device—substrate assemblies **30** are in thermal communication with the cold plate **40**. Accordingly, and during operation of the highly integrated power electronics embedded PCB—cold plate assembly **2**, the power device—substrate assemblies **30** are cooled via the flow of heat from the power device—substrate assemblies **30** to the cold plate **40**. In some variations, the cold plate **40** is formed form an electrically conductive material (e.g., aluminum) and it is desirable that a bonding interface between the power electronics embedded PCB **20** and the cold plate **40** exhibit desired electrical insulation and thermal conduction properties. And in at least one variation, the cold plate **40** includes an internal fluid chamber **41** with an inlet **44**, an outlet **46**, and porous, mesh-structured, machined, or cast heat sinks **43** disposed within the fluid chamber **41**. As used herein, the phrase “bonding interface” refers to one or more layers employed or used to bond a power electronics embedded PCB to a cold plate. And in at least one variation, the phrase “bonding interface” refers to a low thermal resistance dielectric layer employed or used to bond a power electronics embedded PCB to a cold plate.

(22) Still referring to FIG. **2**, a low thermal resistance (LTR) dielectric layer **120** is disposed between the cold plate **40** and the power electronics embedded PCB **20**. For example, in some variations, the LTR dielectric layer **120** is disposed between an upper (+z direction) surface **42** of the cold plate **40** and a lower (−z direction) surface **22** of the power electronics embedded PCB **20**.

(23) The LTR dielectric layer **120** can be formed from any dielectric material and/or dielectric composite material suitable to electrically isolate the power electronics embedded PCB **20** from the cold plate **40**. In some variations, the LTR dielectric layer is a ceramic-polymer composite dielectric material such as BaTiO.sub.3-polymer dielectric materials, Pb(Zr.sub.xTi.sub.1-

x)O.sub.3-polymer dielectric materials (0≤x≤1), and/or SrTiO.sub.3 dielectric materials, among others.
(24) In some variations, the LTR dielectric layers according to the teachings of the present disclosure have an average thickness (z direction) less than about 250 μm, for example less than about 200 μm, less than about 150 μm, less than about 100 μm, less than about 50 μm, less than about 40 μm, less than about 30 μm, less than about 20 μm, or less than about 10 μm. In some variations the LTR dielectric layer has an average thickness between about 10 μm and about 50 μm, for example between about 20 μm and about 40 μm, or between about 25 μm and about 35 μm.
(25) In some variations, the LTR dielectric layer **120** is bonded directly to the upper surface **42** of the cold plate **40**, i.e., the LTR dielectric layer **120** is in direct contact with the upper surface **42**. In at least one variation, the LTR dielectric layer **120** is bonded directly to the lower surface **22** of the power electronics embedded PCB **20**. And in some variations, the LTR dielectric layer **120** is bonded directly to the upper surface **42** of the cold plate **40** and the lower surface **22** of the power electronics embedded PCB **20**. As used herein, the phrase “directly bonded” refers to one component or layer being bonded to and in direct contact with another component or layer, i.e., without any additional components or layers therebetween. And while not shown in the figures, in some variations, a metallization layer can be included on an upper (+z direction) and/or lower (−z direction) surface of the LTR dielectric layer **120**. For example, the cold plate **40** can be formed from an aluminum alloy and an aluminum and/or copper containing metallization layer can be included on the lower surface of the LTR dielectric layer **120**. Also, the power electronics embedded PCB **20** can include a copper containing lower surface **22** and an aluminum and/or copper containing metallization layer can be included on the upper surface of the LTR dielectric layer **120**.
(26) In order to better illustrate the effectiveness of the highly integrated power electronics embedded PCB—cold plate assemblies disclosed herein, but not limit the scope thereof in any manner, results of thermal and mechanical stress simulations for the highly integrated power electronics embedded PCB—cold plate assembly **2** are provided and discussed below.
(27) Referring to FIGS. **3**A-**3**C, results of thermal, normal stress, and shear stress simulations, respectively, for the highly integrated power electronics embedded PCB—cold plate assembly **2** with a LTR dielectric layer **120** bonding interface are shown. In addition, thermal, normal stress, and shear stress simulations for a highly integrated power electronics embedded PCB—cold plate assembly **2** with a grease layer—AlN layer—grease layer bonding interface (not shown) and a highly integrated power electronics embedded PCB—cold plate assembly **2** with a solder layer—AlN layer—solder layer bonding interface (not shown) were performed for comparison purposes.
(28) The highly integrated power electronics embedded PCB—cold plate assembly **2** was modeled as having a power electronics embedded PCB **20** with six (6) SiC MOS power device—substrate assemblies **30** and a bonding interface formed or provided by a LTR dielectric layer **120** with a thermal conductivity of 3 W/m*K and a thickness of about 38 μm. Also, the highly integrated power electronics embedded PCB—cold plate assembly **2** was simulated as operating with six (6) MOSFET power devices and each MOSFET power device had a heat loss level of 286 W.
(29) Regarding thermal simulation of the highly integrated power electronics embedded PCB—cold plate assembly **2** with the LTR dielectric layer **120** bonding interface, the maximum temperature of the SiC MOS power device—substrate assemblies **30** was about 142° C. (FIG. **3**A). In contrast, thermal simulation of the highly integrated power electronics embedded PCB—cold plate assembly **2** with a grease layer—AlN layer—grease layer bonding interface between the power electronics embedded PCB **20** and the cold plate **40** showed a maximum temperature of about 166° C. for the SiC MOS power device—substrate assemblies **30**. Also, thermal simulation of the highly integrated power electronics embedded PCB—cold plate assembly **2** with solder layer—AlN layer—solder layer bonding interface between the power electronics embedded PCB **20** and the cold plate **40** showed a maximum temperature of about 135° C. for the SiC MOS power device

—substrate assemblies **30**. Accordingly, a LTR dielectric layer **120** bonding interface resulted in a maximum temperature for the SiC MOS power device—substrate assemblies **30** 24° C. less than a grease layer—AlN layer—grease layer bonding interface, and only 7° C. higher than a solder layer—AlN layer—solder layer bonding interface. Stated differently, the LTR dielectric layer **120** bonding interface provides desired heat transfer between SiC MOS power device—substrate assemblies embedded in a multi-layer PCB and a cold plate to which the multi-layer PCB, with power device—substrate assemblies embedded therein, is bonded.

(30) Regarding normal stress simulation of the highly integrated power electronics embedded PCB—cold plate assembly **2** with the LTR dielectric layer **120** bonding interface (FIG. **3**B), the average normal stress at the LTR dielectric layer **120** bonding interface was about 2.10×10.sup.5 N/m.sup.2. In contrast, normal stress simulation of the highly integrated power electronics embedded PCB—cold plate assembly **2** with the highly integrated power electronics embedded PCB—cold plate assembly **2** with the solder layer—AlN layer—solder layer bonding interface showed an average normal stress at the solder layer—AlN layer—solder layer bonding interface 5.28×10.sup.5 N/m.sup.2. Accordingly, a LTR dielectric layer **120** bonding interface resulted in a bonding interface normal stress reduction of about 60% compared to a solder layer—AlN layer—solder layer bonding interface.

(31) And regarding shear stress simulation of the highly integrated power electronics embedded PCB—cold plate assembly **2** with the LTR dielectric layer **120** bonding interface (FIG. **3**C), the average shear stress at the LTR dielectric layer **120** bonding interface was about 2.7×10.sup.6 N/m.sup.2. In contrast, shear stress simulation of the highly integrated power electronics embedded PCB—cold plate assembly **2** with the solder layer—AlN layer—solder layer bonding interface showed an average shear stress at the solder layer—AlN layer—solder layer bonding interface 6.0×10.sup.6 N/m.sup.2. Accordingly, a LTR dielectric layer **120** bonding interface resulted in a bonding interface shear stress reduction of about 55% compared to a solder layer—AlN layer—solder layer bonding interface.

(32) Referring now to FIGS. **4**A-**4**E, steps for manufacturing a plurality of highly integrated power electronics embedded PCB—cold plate assemblies **2** are shown. For example, FIG. **4**A illustrates preparing or providing a cold plate fabrication panel **200** from which nine (9) separate cold plates **40** are formed. That is, the cold plate fabrication panel **200** is manufactured to have nine individual cold plates **40** after the cold plate fabrication panel **200** is cut or separated into individual cold plates units. And it should be understood that the cold plate fabrication panel **200** can be manufactured to have less than nine or more than nine individual cold plates **40** after the cold plate fabrication panel **200** is cut or separated into individual cold plates units.

(33) FIG. **4**B illustrates laminating a LTR dielectric layer **120** onto the cold plate fabrication panel **200** and FIG. **4**C illustrates bonding a power electronics embedded PCB fabrication panel **220** to the LTR dielectric layer **120**, and thus to the cold plate fabrication panel **200**. In some variations, the power electronics embedded PCB fabrication panel **220** is bonded to the LTR dielectric layer **120** using heat (i.e., elevated temperature) and pressure (i.e., elevated pressure). For example, in some variations pressures between about 2.1 megapascals (MPa) (300 pounds per square inch (psi)) and about 2.8 MPa (400 psi) are applied to the power electronics embedded PCB fabrication panel **220** and the LTR dielectric layer **120**. In the alternative, or in addition to, the power electronics embedded PCB fabrication panel **220** and the LTR dielectric layer **120** are held at temperatures between about 150° C. (300° F.) and about 204° C. (400° F.).

(34) FIG. **4**D shows a side view of a highly integrated power electronics embedded PCB—cold plate assembly fabrication panel **240** formed from bonding of the power electronics embedded PCB fabrication panel **220** to the cold plate fabrication panel **200** via the LTR dielectric layer **120**. FIG. **4**E illustrates one of the highly integrated power electronics embedded PCB—cold plate assemblies **2** after cutting of the highly integrated power electronics embedded PCB—cold plate assembly fabrication panel **240** into nine separates units. And in some variations an inlet tube **45** is attached

or bonded (e.g., soldered or press fit) to the inlet **44** and an outlet tube **47** is attached or bonded to the outlet **46** of the individual the highly integrated power electronics embedded PCB—cold plate assemblies **2** such that a coolant 'C' flows through the fluid chamber **41** and extracts heat from the power device—substrate assemblies **30** as illustrated in FIG. **4**E.

(35) Referring now to FIG. **5**, a flow chart for a method **50** of manufacturing a plurality of highly integrated power embedded PCB—cold plate assemblies **2** is shown. The method **50** includes preparing a cold plate fabrication panel at **500**, preparing or providing a LTR dielectric layer at **510**, and bonding a power electronics embedded PCB fabrication panel to the cold plate fabrication panel via the LTR dielectric layer to from an integrated power electronics embedded PCB—cold plate fabrication panel at **520**. In some variations, the LTR dielectric layer is laminated onto the cold plate fabrication panel before bonding the bonding the power electronics embedded PCB fabrication panel to the cold plate fabrication panel. In other variations, the LTR dielectric layer is a liquid or film type dielectric material that is applied to the cold plate fabrication or the power electronics embedded PCB fabrication panel before the bonding the bonding the power electronics embedded PCB fabrication panel to the cold plate fabrication panel (e.g., by placing the power electronics embedded PCB fabrication panel and the cold plate fabrication panel with the LTR dielectric layer sandwiched therebetween in an oven for bonding). At **530**, the highly integrated power electronics embedded PCB—cold plate fabrication panel is cut into individual highly integrated power electronics embedded PCB—cold plate assemblies, and in some variations, inlet tubes and outlet tubes are assembled or bonded to the individual highly integrated power electronics embedded PCB—cold plate assemblies at **540**.

(36) The preceding description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or its uses. Work of the presently named inventors, to the extent it may be described in the background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present technology.

(37) The block and/or flow diagrams in the figures illustrate the architecture, functionality, and/or operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the block diagram may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved.

(38) As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A or B or C), using a non-exclusive logical "or." It should be understood that the various steps within a method may be executed in different order without altering the principles of the present disclosure. Disclosure of ranges includes disclosure of all ranges and subdivided ranges within the entire range.

(39) The headings (such as "Background" and "Summary") and sub-headings used herein are intended only for the general organization of topics within the present disclosure and are not intended to limit the disclosure of the technology or any aspect thereof. The recitation of multiple variations or forms having stated features is not intended to exclude other variations or forms having additional features, or other variations or forms incorporating different combinations of the stated features.

(40) As used herein the term "about" when related to numerical values herein refers to known commercial and/or experimental measurement variations or tolerances for the referenced quantity. In some variations, such known commercial and/or experimental measurement tolerances are +/−10% of the measured value, while in other variations such known commercial and/or

experimental measurement tolerances are +/−5% of the measured value, while in still other variations such known commercial and/or experimental measurement tolerances are +/−2.5% of the measured value. And in at least one variation, such known commercial and/or experimental measurement tolerances are +/−1% of the measured value.
(41) The terms “a” and “an,” as used herein, are defined as one or more than one. The term “plurality.” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having.” as used herein, are defined as comprising (i.e., open language). The phrase “at least one of . . . and . . . ” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B, and C” includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC, or ABC).
(42) As used herein, the terms “comprise” and “include” and their variants are intended to be non-limiting, such that recitation of items in succession or a list is not to the exclusion of other like items that may also be useful in the devices and methods of this technology. Similarly, the terms “can” and “may” and their variants are intended to be non-limiting, such that recitation that a form or variation can or may comprise certain elements or features does not exclude other forms or variations of the present technology that do not contain those elements or features.
(43) The broad teachings of the present disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent to the skilled practitioner upon a study of the specification and the following claims. Reference herein to one variation, or various variations means that a particular feature, structure, or characteristic described in connection with a form or variation or particular system is included in at least one variation or form. The appearances of the phrase “in one variation” (or variations thereof) are not necessarily referring to the same variation or form. It should also be understood that the various method steps discussed herein do not have to be conducted in the same order as depicted, and not each method step is required in each variation or form.
(44) The foregoing description of the forms and variations has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular form or variation are generally not limited to that particular form or variation, but, where applicable, are interchangeable and can be used in a selected form or variation, even if not specifically shown or described. The same may also be varied in many ways. Such variations should not be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. An assembly comprising: a cold plate; a power electronics embedded printed circuit board (PCB); and a low thermal resistance (LTR) dielectric layer in direct contact with and directly bonded to the cold plate, the LTR dielectric layer sandwiched between the cold plate and the power electronics embedded PCB such that the power electronics embedded PCB is bonded to the cold plate via the LTR dielectric layer to form a highly integrated power electronics embedded PCB—cold plate assembly.
2. The assembly according to claim 1, wherein the LTR dielectric layer is a ceramic-polymer composite dielectric material.
3. The assembly according to claim 2, wherein the ceramic-polymer composite dielectric material selected from the group consisting of BaTiO.sub.3-polymer dielectric materials, Pb(Zr.sub.xTi.sub.1-x)O.sub.3-polymer dielectric materials (0≤x≤1), and SrTiO.sub.3 dielectric materials.
4. The assembly according to claim 2, wherein the ceramic-polymer composite dielectric material

selected is a $BaTiO_3$-polymer dielectric material.
5. The assembly according to claim 2, wherein the ceramic-polymer composite dielectric material selected is a $Pb(Zr_xTi_{1-x})O_3$-polymer dielectric materials ($0 \le x \le 1$).
6. The assembly according to claim 2, wherein the ceramic-polymer composite dielectric material is a $SrTiO_3$ dielectric material.
7. The assembly according to claim 1, wherein the LTR dielectric layer is directly bonded to the power electronics embedded PCB.
8. The assembly according to claim 1, wherein the LTR dielectric layer has an average thickness between about 30 μm and 50 μm.
9. The assembly according to claim 8, wherein the average thickness is between about 35 μm and 45 μm.
10. The assembly according to claim 1, wherein the cold plate is formed from an aluminum alloy.
11. The assembly according to claim 1, wherein the cold plate comprises a fluid chamber configured for a coolant to flow therethrough.
12. The assembly according to claim 11 further comprising porous, mesh-structured, machined, or cast heat sinks disposed within the fluid chamber.
13. A highly integrated power electronics embedded printed circuit board (PCB)—cold plate assembly comprising: a cold plate; a power electronics embedded PCB; and a low thermal resistance dielectric (LTR) layer in direct contact with and directly bonded to the cold plate, the LTR dielectric layer sandwiched between and directly bonded to the cold plate and the power electronics embedded PCB.
14. The highly integrated power electronics embedded PCB—cold plate assembly according to claim 13, wherein the LTR dielectric layer has an average thickness between about 30 μm and 50 μm.
15. The highly integrated power electronics embedded PCB—cold plate assembly according to claim 13, wherein the LTR dielectric layer is a ceramic-polymer composite dielectric material selected from the group consisting of $BaTiO_3$-polymer dielectric materials, $Pb(Zr_xTi_{1-x})O_3$-polymer dielectric materials ($0 \le x \le 1$), and $SrTiO_3$ dielectric materials.
16. A method comprising: laminating a low thermal resistance (LTR) dielectric layer onto a cold plate fabrication panel such that the LTR dielectric layer is in directly bonded to the cold plate fabrication panel, the cold plate fabrication panel comprising a plurality of individual cold plate units; bonding a power electronics embedded printed circuit board (PCB) fabrication panel to the LTR dielectric layer using elevated temperature and elevated pressure and forming an integrated power electronics PCB—cold plate fabrication panel; and cutting the integrated power electronics PCB—cold plate fabrication panel into a plurality of individual highly integrated power electronics embedded PCB—cold plate units.
17. The method according to claim 16 further comprising assembling an inlet tube and an outlet tube on the plurality of individual highly integrated power electronics embedded PCB—cold plate units.
18. The method according to claim 16, wherein the LTR dielectric layer is directly bonded to the power electronics embedded PCB fabrication panel.
19. The method according to claim 16, wherein the LTR dielectric layer has an average thickness between about 30 μm and 50 μm.
20. The method according to claim 16, wherein the cold plate fabrication panel is formed from an aluminum alloy, the power electronics embedded PCB fabrication panel comprises a copper containing layer in direct contact with the LTR dielectric layer.